

PHYS 121 — HW4 Solutions

WRITTER

November 14, 2025

Q1. A rotating grinding wheel (10 pts)

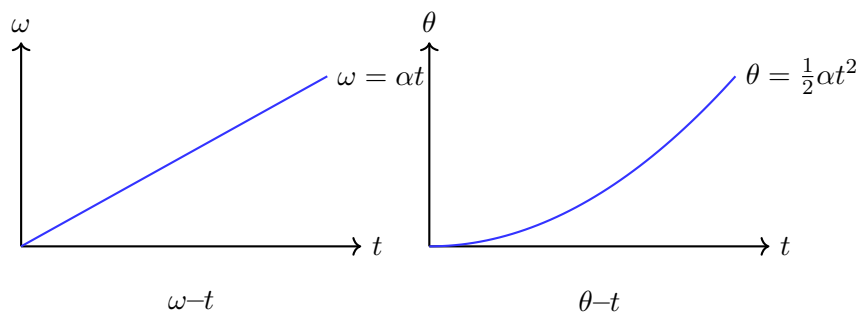
Given: $M = 50.0$ kg, diameter $D = 1.0$ m, so radius $R = 0.50$ m.

(a) **Spin-up torque:** From rest ($\omega_0 = 0$) to $\omega = 120$ rev/min $= 4\pi$ rad/s $= 12.6$ rad/s in $\Delta\theta = 20$ revolutions $= 40\pi$ rad $= 125.7$ rad.

Using rotational kinematics: $\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$:

$$\omega^2 = 2\alpha\Delta\theta \Rightarrow \alpha = \frac{\omega^2}{2\Delta\theta} = \frac{(4\pi)^2}{2 \times 40\pi} = \frac{16\pi^2}{80\pi} = \frac{\pi}{5} = \boxed{0.628 \text{ rad/s}^2}.$$

Sketches:

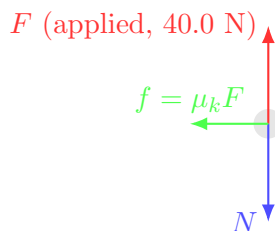


Moment of inertia for a cylinder: $I = \frac{1}{2}MR^2 = \frac{1}{2} \times 50.0 \times (0.50)^2 = 6.25 \text{ kg} \cdot \text{m}^2$.

Torque: $\tau = I\alpha = 6.25 \times 0.628 = \boxed{3.93 \text{ N} \cdot \text{m}}$.

(b) **Steady-speed torque:** Tool pressed perpendicularly at rim with $F = 40.0$ N, $\mu_k = 0.60$.

FBD at contact point:



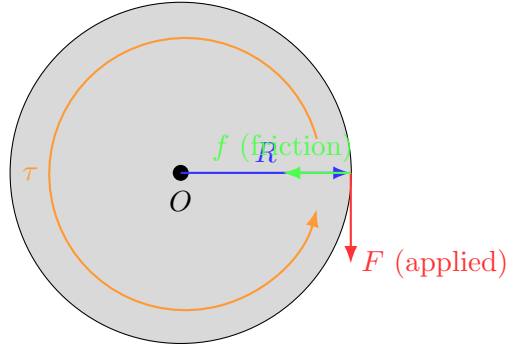
Contact point FBD

Friction force: $f = \mu_k F = 0.60 \times 40.0 = 24.0$ N (tangential, opposing rotation).

Friction torque (opposing rotation): $\tau_{\text{friction}} = fR = 24.0 \times 0.50 = \boxed{12.0 \text{ N} \cdot \text{m}}$.

To maintain constant ω (zero angular acceleration), the motor must provide equal and opposite torque: $\tau_{\text{motor}} = \tau_{\text{friction}} = \boxed{12.0 \text{ N} \cdot \text{m}}$.

Top view



Q2. Two-groove Atwood pulley (10 pts)

Given: Pulley with moment of inertia I , two radii r_1 and r_2 ($r_2 > r_1$), masses m_1 and m_2 ($m_2 > m_1$).

Choose positive rotation so that m_2 moves down (clockwise).

(a) Angular acceleration: Tensions T_1 and T_2 act on the pulley. For translation of masses:

$$m_1 a_1 = T_1 - m_1 g, \quad m_2 a_2 = m_2 g - T_2.$$

No-slip condition: $a_1 = \alpha r_1$, $a_2 = \alpha r_2$.

Net torque on pulley: $\tau_{\text{net}} = T_2 r_2 - T_1 r_1 = I \alpha$.

Solving:

$$T_1 = m_1(g + a_1) = m_1(g + \alpha r_1),$$

$$T_2 = m_2(g - a_2) = m_2(g - \alpha r_2),$$

$$T_2 r_2 - T_1 r_1 = I \alpha.$$

Substituting:

$$m_2(g - \alpha r_2)r_2 - m_1(g + \alpha r_1)r_1 = I \alpha.$$

$$m_2 g r_2 - m_1 g r_1 - \alpha(m_2 r_2^2 + m_1 r_1^2) = I \alpha.$$

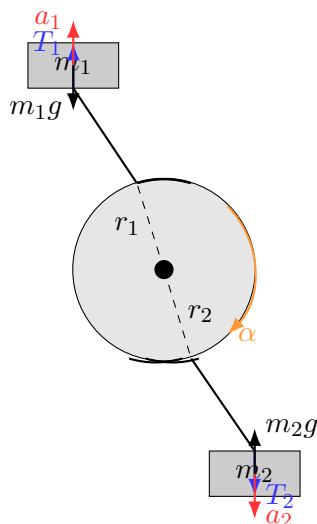
$$m_2 g r_2 - m_1 g r_1 = \alpha(I + m_2 r_2^2 + m_1 r_1^2).$$

$$\alpha = \boxed{\frac{m_2 g r_2 - m_1 g r_1}{I + m_2 r_2^2 + m_1 r_1^2}}.$$

(b) Linear accelerations:

$$a_1 = \alpha r_1 = \boxed{\frac{(m_2 g r_2 - m_1 g r_1)r_1}{I + m_2 r_2^2 + m_1 r_1^2}} \quad (\text{upward}),$$

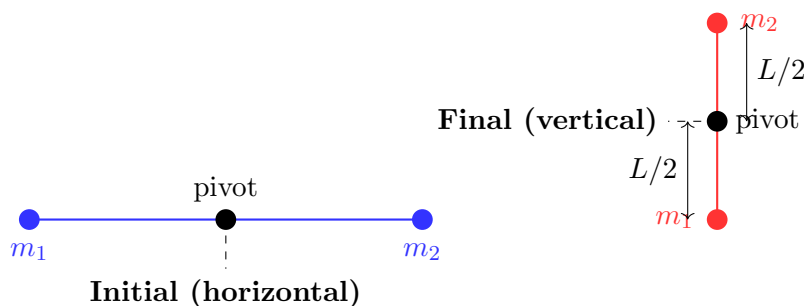
$$a_2 = \alpha r_2 = \boxed{\frac{(m_2 g r_2 - m_1 g r_1)r_2}{I + m_2 r_2^2 + m_1 r_1^2}} \quad (\text{downward}).$$



Q3. Rotating stick with end masses (10 pts)

Given: Stick length $L = 1.0$ m, mass $M = 6.0$ kg, pivots about center. Masses $m_1 = 4.0$ kg and $m_2 = 2.5$ kg at ends. Released from horizontal.

(a) Drawing:



(b) **Energy method:** Only the end masses change gravitational energy. The stick's center of mass remains at the pivot, so $\Delta U_{\text{stick}} = 0$.

Initial: Both masses at height $h_0 = 0$ (taking pivot as reference).

Final: m_1 at height $-L/2 = -0.50$ m, m_2 at height $L/2 = 0.50$ m.

$$\Delta U = m_1 g (-L/2) + m_2 g (L/2) = g(m_2 - m_1)L/2 = 9.8 \times (2.5 - 4.0) \times 0.50 = \boxed{-7.35 \text{ J}}.$$

The negative sign indicates energy is released (converted to kinetic energy).

(c) **Rotational inertia and angular speed:** Rotational inertia about the pivot:

$$I = I_{\text{stick}} + m_1(L/2)^2 + m_2(L/2)^2.$$

For a stick pivoted at center: $I_{\text{stick}} = \frac{1}{12}ML^2$.

$$I = \frac{1}{12} \times 6.0 \times 1.0^2 + 4.0 \times 0.50^2 + 2.5 \times 0.50^2 = 0.50 + 1.00 + 0.625 = 2.125 \text{ kg} \cdot \text{m}^2.$$

Conservation of energy: $\Delta K + \Delta U = 0$, so $K_{\text{final}} = -\Delta U = 7.35 \text{ J}$.

$$\frac{1}{2}I\omega^2 = 7.35 \quad \Rightarrow \quad \omega = \sqrt{\frac{2 \times 7.35}{2.125}} = \sqrt{6.918} = \boxed{2.63 \text{ rad/s}}.$$

(d) Checks:

- If $m_1 = m_2$: $\Delta U = g(m_2 - m_1)L/2 = 0$, so $\omega = 0$ (no rotation, balanced).
- If $m_2 \rightarrow 0$: Only m_1 at distance $L/2$ from pivot. $I = \frac{1}{12}ML^2 + m_1(L/2)^2$, $\Delta U = -m_1gL/2$, giving $\omega = \sqrt{m_1gL/I}$. This matches a single point mass at $L/2$ if the stick is massless ($M = 0$), in which case $I = m_1(L/2)^2$ and $\omega = \sqrt{2g/L}$.

Q4. Spring-pushed sphere that skids then rolls (17 pts)

Given: Solid ball mass m , radius r , spring constant k , compression x . Initially at rest on level surface.

(a) Initial translational velocity: Energy stored in spring: $U_{\text{spring}} = \frac{1}{2}kx^2$.

Assuming ball is not rotating and no friction, all energy goes to translational kinetic energy:

$$\frac{1}{2}mv_i^2 = \frac{1}{2}kx^2 \quad \Rightarrow \quad v_i = \sqrt{\frac{kx^2}{m}} = x\sqrt{\frac{k}{m}}.$$

(b) Angular momentum about contact point: Initially, ball is not rotating, so angular momentum about contact point is:

$$L = mv_i r = \boxed{mrx\sqrt{\frac{k}{m}} = xr\sqrt{km}}.$$

(c) Moment of inertia about contact point: Using parallel axis theorem. For a solid sphere about its center: $I_{\text{cm}} = \frac{2}{5}mr^2$.

$$\text{About the contact point: } I = I_{\text{cm}} + mr^2 = \frac{2}{5}mr^2 + mr^2 = \boxed{\frac{7}{5}mr^2}.$$

(d) Final velocity from angular momentum conservation: When rolling without slipping, $v_f = \omega r$, so $\omega = v_f/r$.

$$\text{Angular momentum about contact point (conserved): } L = I\omega = \frac{7}{5}mr^2 \cdot \frac{v_f}{r} = \frac{7}{5}mr v_f.$$

$$\text{Conservation: } mv_i r = \frac{7}{5}mr v_f, \text{ so } v_f = \frac{5}{7}v_i = \boxed{\frac{5}{7}x\sqrt{\frac{k}{m}}}.$$

(e) Energy dissipated: Initial energy: $E_i = \frac{1}{2}kx^2$.

$$\text{Final energy (rolling): } E_f = \frac{1}{2}mv_f^2 + \frac{1}{2}I_{\text{cm}}\omega^2 = \frac{1}{2}mv_f^2 + \frac{1}{2} \cdot \frac{2}{5}mr^2 \cdot \frac{v_f^2}{r^2} = \frac{1}{2}mv_f^2 + \frac{1}{5}mv_f^2 = \frac{7}{10}mv_f^2.$$

$$\text{Using } v_f = \frac{5}{7}v_i \text{ and } v_i^2 = \frac{kx^2}{m}:$$

$$E_f = \frac{7}{10}m \cdot \frac{25}{49}v_i^2 = \frac{7}{10} \cdot \frac{25}{49} \cdot kx^2 = \frac{5}{14}kx^2.$$

$$\text{Energy dissipated: } \Delta E = E_i - E_f = \frac{1}{2}kx^2 - \frac{5}{14}kx^2 = \frac{7-5}{14}kx^2 = \frac{2}{14}kx^2 = \frac{1}{7}kx^2.$$

$$\text{Percentage: } \frac{\Delta E}{E_i} \times 100\% = \frac{1/7}{1/2} \times 100\% = \frac{2}{7} \times 100\% = \boxed{28.6\%}.$$

(f) Toy estimate: $F = 10 \text{ N}$ to compress $x = 1 \text{ cm} = 0.01 \text{ m}$, so $k = F/x = 1000 \text{ N/m}$.

Ball size of grape: mass $m \approx 5 \text{ g} = 0.005 \text{ kg}$, radius $r \approx 1 \text{ cm} = 0.01 \text{ m}$.

$$v_f = \frac{5}{7} \times 0.01 \times \sqrt{\frac{1000}{0.005}} = \frac{5}{7} \times 0.01 \times \sqrt{200000} = \frac{5}{7} \times 0.01 \times 447.2 = \boxed{3.19 \text{ m/s}}.$$

Q5. Paper roll on an accelerating truck (10 pts)

Given: Solid cylindrical roll on flat truck bed. Truck accelerates forward at constant a .

1) **Newton's 2nd law:** In ground frame, for translation:

$$ma_{\text{cm}} = f,$$

where f is the friction force (forward, driving the roll).

For rotation about center of mass:

$$I\alpha = fR,$$

where $I = \frac{1}{2}mR^2$ for solid cylinder, and f provides torque.

2) **No-slip condition:** Acceleration of contact point equals truck's acceleration:

$$a_{\text{contact}} = a_{\text{cm}} + R\alpha = a.$$

From rotational equation: $\alpha = \frac{fR}{I} = \frac{fR}{\frac{1}{2}mR^2} = \frac{2f}{mR}$.

Substituting: $a_{\text{cm}} + R \cdot \frac{2f}{mR} = a_{\text{cm}} + \frac{2f}{m} = a$.

From translational equation: $f = ma_{\text{cm}}$, so:

$$a_{\text{cm}} + \frac{2ma_{\text{cm}}}{m} = 3a_{\text{cm}} = a \quad \Rightarrow \quad a_{\text{cm}} = \boxed{\frac{a}{3}}.$$

3) **Relative acceleration:** The roll's acceleration relative to the truck:

$$a_{\text{rel}} = a_{\text{cm}} - a = \frac{a}{3} - a = \boxed{-\frac{2a}{3}}.$$

Negative sign indicates the roll accelerates backward relative to the truck.

4) **Distance to drop:** If the roll starts at the front edge and the truck bed has length L , the roll reaches the back edge when:

$$s_{\text{truck}} = \frac{1}{2}at^2, \quad s_{\text{roll}} = \frac{1}{2}a_{\text{cm}}t^2 = \frac{1}{2} \cdot \frac{a}{3}t^2 = \frac{at^2}{6}.$$

Relative distance: $s_{\text{rel}} = s_{\text{truck}} - s_{\text{roll}} = \frac{at^2}{2} - \frac{at^2}{6} = \frac{at^2}{3}$.

For the roll to reach the back (relative distance = L):

$$L = \frac{at^2}{3} \quad \Rightarrow \quad t^2 = \frac{3L}{a}.$$

$$\text{Distance truck moves: } s = \frac{1}{2}at^2 = \frac{1}{2}a \cdot \frac{3L}{a} = \boxed{\frac{3L}{2}}.$$

Q6. Bug hits a hanging stick (10 pts)

Given: Bug (mass m_b , speed u) hits and sticks to end of uniform stick (length L , mass $M = 10m_b$), pivoted at top, hanging vertically.

1) **Angular momentum conservation:** About the pivot, before impact: bug has angular momentum $L_i = m_b u L$ (perpendicular distance is L).

After impact: stick + bug rotate with angular speed ω . Rotational inertia about pivot:

$$I = I_{\text{stick}} + m_b L^2 = \frac{1}{3} M L^2 + m_b L^2 = \frac{1}{3} \cdot 10 m_b L^2 + m_b L^2 = \frac{10 m_b L^2}{3} + m_b L^2 = \frac{13 m_b L^2}{3}.$$

Conservation: $m_b u L = I \omega = \frac{13 m_b L^2}{3} \omega$.

$$\omega = \frac{3 m_b u L}{13 m_b L^2} = \boxed{\frac{3u}{13L}}.$$

2) Maximum angle: After impact, energy is conserved. Initial kinetic energy:

$$K_i = \frac{1}{2} I \omega^2 = \frac{1}{2} \cdot \frac{13 m_b L^2}{3} \cdot \left(\frac{3u}{13L} \right)^2 = \frac{13 m_b L^2}{6} \cdot \frac{9u^2}{169 L^2} = \frac{3 m_b u^2}{26}.$$

At maximum angle θ , the center of mass rises. The stick's center of mass is at $L/2$ from pivot, bug is at L from pivot.

Initial center of mass distance from pivot: $d_{\text{cm}} = \frac{M \cdot L/2 + m_b \cdot L}{M + m_b} = \frac{10 m_b \cdot L/2 + m_b L}{11 m_b} = \frac{6 m_b L}{11 m_b} = \frac{6L}{11}$.

Taking the pivot as the reference (height $h = 0$), initially the center of mass is at height $h_i = -\frac{6L}{11}$ (below the pivot).

At maximum angle θ from vertical, the center of mass is at height $h_f = -\frac{6L}{11} \cos \theta$.

Rise: $\Delta h = h_f - h_i = -\frac{6L}{11} \cos \theta - \left(-\frac{6L}{11} \right) = \frac{6L}{11} (1 - \cos \theta)$.

Energy conservation: $K_i = (M + m_b) g \Delta h = 11 m_b g \cdot \frac{6L}{11} (1 - \cos \theta) = 6 m_b g L (1 - \cos \theta)$.

$$\frac{3 m_b u^2}{26} = 6 m_b g L (1 - \cos \theta) \quad \Rightarrow \quad 1 - \cos \theta = \frac{u^2}{52 g L}.$$

$$\cos \theta = 1 - \frac{u^2}{52 g L}.$$

Solving for L :

$$\frac{u^2}{52 g L} = 1 - \cos \theta \quad \Rightarrow \quad L = \boxed{\frac{u^2}{52 g (1 - \cos \theta)}}.$$

3) Check: If $\theta \rightarrow 0$, then $1 - \cos \theta \rightarrow 0$, so $L \rightarrow \infty$. This makes sense: to achieve only a tiny swing, the stick must be very long (small angular speed, large moment of inertia).

Q7. Ice-skating twirl and the double Axel (15 pts)

Given: Skater as solid cylinder: height $h = 1.50$ m, radius $R = 10.0$ cm = 0.10 m, mass $M = 45$ kg. Two arms: each length $\ell = 0.60$ m, mass $m_a = 2.4$ kg. Arms can be extended or against cylinder.

(a) Rotational inertia:

i) Arms at side: Arms treated as thin rods against cylinder surface. For a thin rod of length ℓ and mass m about one end: $I = \frac{1}{3} m \ell^2$. But if held against cylinder, they contribute as point masses at distance R from axis: $I_{\text{arms}} = 2 \times m_a R^2 = 2 \times 2.4 \times 0.10^2 = 0.048 \text{ kg} \cdot \text{m}^2$.

Actually, if arms are against the cylinder, their center of mass is at R from axis, so: $I_{\text{arms}} = 2 \times m_a R^2 = 0.048 \text{ kg} \cdot \text{m}^2$.

Cylinder: $I_{\text{cylinder}} = \frac{1}{2} M R^2 = \frac{1}{2} \times 45 \times 0.10^2 = 0.225 \text{ kg} \cdot \text{m}^2$.

Total: $I_1 = 0.225 + 0.048 = \boxed{0.273 \text{ kg} \cdot \text{m}^2}$.

ii) Arms extended: Each arm extends perpendicular from the cylinder surface. If the arm is a thin rod of length ℓ with its inner end at distance R from the axis, the center of mass is at distance $R + \ell/2 = 0.10 + 0.30 = 0.40$ m from the axis.

Using parallel axis theorem: $I_{\text{arm}} = I_{\text{cm}} + m_a d^2$, where $I_{\text{cm}} = \frac{1}{12} m_a \ell^2$ (rod about its center) and $d = R + \ell/2 = 0.40$ m.

$$I_{\text{arm}} = \frac{1}{12} m_a \ell^2 + m_a (R + \ell/2)^2 = \frac{1}{12} \times 2.4 \times 0.60^2 + 2.4 \times 0.40^2 = 0.072 + 0.384 = 0.456 \text{ kg} \cdot \text{m}^2.$$

For two arms: $I_{\text{arms}} = 2 \times 0.456 = 0.912 \text{ kg} \cdot \text{m}^2$.

Total: $I_2 = 0.225 + 0.912 = \boxed{1.137 \text{ kg} \cdot \text{m}^2}$.

(b) Time in air: Maximum height $h_{\text{max}} = 0.9$ m. Using $v^2 = v_0^2 - 2gh$, at maximum height $v = 0$:

$$0 = v_0^2 - 2gh_{\text{max}} \Rightarrow v_0 = \sqrt{2gh_{\text{max}}} = \sqrt{2 \times 9.8 \times 0.9} = \sqrt{17.64} = 4.2 \text{ m/s}.$$

Time to reach maximum: $t_{\text{up}} = v_0/g = 4.2/9.8 = 0.429$ s.

Total time in air: $T = 2t_{\text{up}} = \boxed{0.86 \text{ s}}$.

(c) Necessary rotational speed: For 2.5 rotations in time $T = 0.86$ s:

$$\theta = 2.5 \times 2\pi = 5\pi \text{ rad}.$$

$$\omega = \frac{\theta}{T} = \frac{5\pi}{0.86} = \boxed{18.3 \text{ rad/s} = 175 \text{ rev/min}}.$$

(d) Configuration and initial speed: To achieve high ω , start with arms extended (larger I) and small ω_i , then pull arms in (smaller I) to increase ω via angular momentum conservation.

Angular momentum conservation: $I_2 \omega_i = I_1 \omega_f$, where $\omega_f = 18.3 \text{ rad/s}$.

$$\omega_i = \frac{I_1}{I_2} \omega_f = \frac{0.273}{1.137} \times 18.3 = 0.240 \times 18.3 = \boxed{4.39 \text{ rad/s}}.$$

So: start with arms extended, $\omega_i = 4.39 \text{ rad/s}$, then quickly pull arms in to achieve $\omega_f = 18.3 \text{ rad/s}$.

(e) Landing with one leg extended: Extending one leg increases moment of inertia, which decreases angular speed via conservation of angular momentum. This allows better control and stability upon landing, preventing over-rotation.

Q8. Three-force balance on a massless rod (10 pts)

Given: Massless rod with three perpendicular forces at points A, B, C. Rod in static equilibrium. From the figure: point A is at 0.3 m from left end, point B at 0.7 m from left end, point C at 10.4 m from left end. Force $F_C = 100 \text{ N}$ downward at point C.

For static equilibrium: $\sum \vec{F} = 0$ and $\sum \tau = 0$ (about any point).

Taking upward as positive, force balance:

$$F_A + F_B + F_C = 0 \Rightarrow F_A + F_B - 100 = 0 \Rightarrow F_A + F_B = 100 \text{ N}.$$

Taking torques about point A (where F_A acts, so its torque is zero). Distances: $d_{AB} = 0.7 - 0.3 = 0.4$ m, $d_{AC} = 10.4 - 0.3 = 10.1$ m.

Torque balance about A (taking counterclockwise as positive). Since F_C is downward and acts to the right of A, it creates a clockwise (negative) torque. F_B must create a counterclockwise (positive) torque, so it must be upward:

$$F_B \cdot d_{AB} - F_C \cdot d_{AC} = 0.$$

With $F_C = 100 \text{ N}$ (downward, magnitude) and distances in meters:

$$F_B \times 0.4 - 100 \times 10.1 = 0 \Rightarrow 0.4F_B = 1010 \Rightarrow F_B = \frac{1010}{0.4} = \boxed{2525 \text{ N}} \text{ (upward).}$$

From force balance (taking upward as positive): $F_A + F_B + F_C = 0$, so $F_A + 2525 - 100 = 0$, giving:

$$F_A = 100 - 2525 = \boxed{-2425 \text{ N}} = 2425 \text{ N downward.}$$

Verification: Taking torques about point B (taking counterclockwise as positive): - F_A (downward, left of B): creates clockwise torque $= -|F_A| \times d_{AB} = -2425 \times 0.4 = -970 \text{ N}\cdot\text{m}$ - F_C (downward, right of B): creates counterclockwise torque $= +|F_C| \times d_{BC} = 100 \times 9.7 = 970 \text{ N}\cdot\text{m}$ - Net torque: $-970 + 970 = 0$ (verified).

Taking torques about point C (taking counterclockwise as positive): - F_A (downward, left of C): creates counterclockwise torque $= -F_A \times d_{AC} = -(-2425) \times 10.1 = 24492.5 \text{ N}\cdot\text{m}$ - F_B (upward, left of C): creates clockwise torque $= -F_B \times d_{BC} = -2525 \times 9.7 = -24492.5 \text{ N}\cdot\text{m}$ - Net torque: $24492.5 - 24492.5 = 0$ (verified).

Summary:

- $F_A = \boxed{2425 \text{ N}}$ downward
- $F_B = \boxed{2525 \text{ N}}$ upward
- $F_C = 100 \text{ N}$ downward (given)

Q9. Spider silk realism + Spiderman (16 pts)

Given: Elastic modulus $Y = 25 \text{ GPa} = 2.5 \times 10^{10} \text{ N/m}^2$. For silk thread: length L , cross-sectional area A , stretch x , tension $F = YAx/L$.

Spider: mass $m = 50 \text{ mg} = 5 \times 10^{-5} \text{ kg}$, line diameter $d = 0.14 \text{ mm} = 1.4 \times 10^{-4} \text{ m}$, length $L = 50 \text{ cm} = 0.50 \text{ m}$.

(a) Stretch: Cross-sectional area: $A = \pi(d/2)^2 = \pi \times (7 \times 10^{-5})^2 = \pi \times 4.9 \times 10^{-9} = 1.54 \times 10^{-8} \text{ m}^2$.

Tension equals weight: $F = mg = 5 \times 10^{-5} \times 9.8 = 4.9 \times 10^{-4} \text{ N}$.

$$F = \frac{YAx}{L} \Rightarrow x = \frac{FL}{YA}.$$

$$x = \frac{4.9 \times 10^{-4} \times 0.50}{2.5 \times 10^{10} \times 1.54 \times 10^{-8}} = \frac{2.45 \times 10^{-4}}{3.85 \times 10^2} = \boxed{6.36 \times 10^{-7} \text{ m} = 0.636 \mu\text{m}}.$$

(b) Potential energy: $U = \frac{1}{2}kx^2$, where effective spring constant $k = YA/L$.

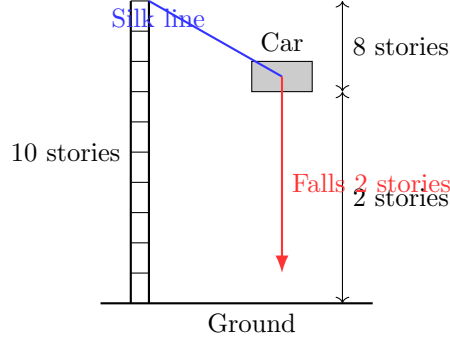
$$k = \frac{2.5 \times 10^{10} \times 1.54 \times 10^{-8}}{0.50} = 7.7 \times 10^2 \text{ N/m}.$$

$$U = \frac{1}{2} \times 770 \times (6.36 \times 10^{-7})^2 = \frac{1}{2} \times 770 \times 4.04 \times 10^{-13} = \boxed{1.56 \times 10^{-10} \text{ J}}.$$

(c) **Speed:** If all energy converts to kinetic: $\frac{1}{2}mv^2 = U$.

$$v = \sqrt{\frac{2U}{m}} = \sqrt{\frac{2 \times 1.56 \times 10^{-10}}{5 \times 10^{-5}}} = \sqrt{6.24 \times 10^{-6}} = \boxed{2.50 \times 10^{-3} \text{ m/s} = 2.50 \text{ mm/s}}.$$

(d) **Drawing:**



(e) **Minimum diameter (infinitely elastic):** Car falls 2 stories. Assume story height ≈ 3 m, so fall height from connection point to ground $h = 6$ m. Car mass $m_{\text{car}} \approx 1500$ kg.

Car falls from top (10 stories) and line connects at 2 stories above ground. Distance fallen: $h = 8 \times 3 \text{ m} = 24$ m. Speed at connection: $v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 24} = \sqrt{470.4} = 21.7 \text{ m/s}$.

Car must stop at ground (additional 2 stories = 6 m). Kinetic energy to absorb: $K = \frac{1}{2}mv^2 = \frac{1}{2} \times 1500 \times (21.7)^2 = 3.53 \times 10^5 \text{ J}$.

If the line is infinitely elastic and stretches by $x = 6$ m (the distance car falls after connection), the energy stored is $U = \frac{1}{2}kx^2$, where $k = YA/L_0$.

Assuming the line's unstretched length is approximately the connection height ($L_0 \approx 6$ m), then:

$$U = \frac{1}{2} \cdot \frac{YA}{L_0} \cdot x^2 = \frac{1}{2} \cdot \frac{YA}{6} \cdot 36 = 3YA.$$

Setting $U = K = 3.53 \times 10^5 \text{ J}$:

$$3 \times 2.5 \times 10^{10} \times A = 3.53 \times 10^5 \quad \Rightarrow \quad A = \frac{3.53 \times 10^5}{7.5 \times 10^{10}} = 4.71 \times 10^{-6} \text{ m}^2.$$

$$d = 2\sqrt{\frac{A}{\pi}} = 2\sqrt{\frac{4.71 \times 10^{-6}}{\pi}} = 2\sqrt{1.50 \times 10^{-6}} = 2 \times 1.225 \times 10^{-3} = \boxed{2.45 \times 10^{-3} \text{ m} = 2.45 \text{ mm}}.$$

(f) **Would the line break?** Real silk has a breaking stress. Typical spider silk breaking stress $\sigma_{\text{max}} \approx 1 \text{ GPa} = 1 \times 10^9 \text{ Pa}$ (dragline silk can be higher, up to $\sim 1.3 \text{ GPa}$, but we'll use 1 GPa).

Maximum tension: $F_{\text{max}} = \sigma_{\text{max}}A = 1 \times 10^9 \times 4.71 \times 10^{-6} = 4.71 \times 10^3 \text{ N} = 4710 \text{ N}$.

The tension in the line when stretched by x is $F = kx = \frac{YA}{L_0}x$. For $x = 6$ m and $L_0 = 6$ m:

$$F = \frac{2.5 \times 10^{10} \times 4.71 \times 10^{-6}}{6} \times 6 = 2.5 \times 10^{10} \times 4.71 \times 10^{-6} = 1.18 \times 10^5 \text{ N}.$$

This far exceeds $F_{\text{max}} = 4.71 \times 10^3 \text{ N}$, so *thelinewouldbreak*.

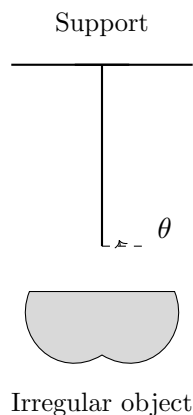
Even if we assume the line can stretch more (making k smaller), the tension required to stop the car would still be very large. For a constant deceleration, if the car stops over distance $x = 6$ m with initial speed $v = 21.7$ m/s:

$$v^2 = 2ax \quad \Rightarrow \quad a = \frac{v^2}{2x} = \frac{(21.7)^2}{12} = 39.2 \text{ m/s}^2 = 4g.$$

Force needed: $F = ma = 1500 \times 39.2 = 5.88 \times 10^4$ N, which is much larger than F_{max} . The silk cannot provide enough force to stop the car safely without exceeding its breaking stress.

Bonus. Measure a moment of inertia (0–2 pts)

A torsion pendulum can measure the moment of inertia of an irregular object. The object is suspended by a wire or fiber that twists when the object rotates. The restoring torque is $\tau = -\kappa\theta$, where κ is the torsion constant and θ is the angular displacement. This gives simple harmonic motion with period $T = 2\pi\sqrt{I/\kappa}$, where I is the total moment of inertia. By measuring T with and without the object, and knowing the wire's κ (from calibration with a known I), one can determine the object's moment of inertia. The setup requires a rigid suspension, minimal air resistance, and precise timing of oscillations.



Assistance notes:

LLMs were used as translators and concept explainers. No direct solutions were provided.